

A deterministic column-balanced greedy construction for the column-descent-free SYT existence conjecture

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Abstract

Let $\widehat{\lambda} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_\ell)$ be a partition with $\ell \geq 2$ and $\widehat{\lambda}_i \geq 2$ for all i . Let $q \geq 1$ and put $\lambda = (\widehat{\lambda}, 1^q)$. Assume the threshold condition $\tau(\widehat{\lambda}) := \max(0, q + 2\ell - 1 - |\widehat{\lambda}|) = 0$, equivalently $q \leq |\widehat{\lambda}| - 2\ell + 1$. Conjecture 4.6 of the converse-per-SYT-positivity paper asserts that there exists a standard Young tableau T^* of shape λ such that (i) $T^*(1, 1) = 1$, $T^*(1, 2) = 2$ and (ii) no j has $j + 1$ directly below j (no column descent). This conjecture was previously known only by brute force enumeration up to $|\lambda| \leq 14$.

We exhibit an explicit deterministic algorithm, the *column-balanced greedy* (Algorithm N), which on every input satisfying the hypotheses produces a candidate T^* . We prove that Algorithm N produces a valid SYT and verify computationally that on every $\tau = 0$ shape with $|\lambda| \leq 20$ (1525 shapes), Algorithm N's output satisfies (i) and (ii). We isolate the remaining structural gap as a clean invariant about the column-fullness counters, which we conjecture but do not prove in full generality.

This gives a constructive description of T^* that subsumes both the “interleaved T^* ” of Definition 4.7 and the manual examples used to close the multi-length-3-row gap in Proposition 4.8.

1 Setup

Throughout, $\widehat{\lambda} = (\widehat{\lambda}_1, \dots, \widehat{\lambda}_\ell)$ is a partition with $\ell \geq 2$ rows, each of length at least 2. Set $L = \ell$, $M = |\widehat{\lambda}| - 2L$ (the number of cells of $\widehat{\lambda}$ in columns ≥ 3), and $\lambda = (\widehat{\lambda}, 1^q)$ for some integer $q \geq 1$. Throughout we work in 0-indexed coordinates: row 0 is the top row, column 0 is the leftmost column. We write a cell as (r, c) where r is the 0-indexed row and c is the 0-indexed column.

The threshold condition $\tau(\widehat{\lambda}) = 0$ is

$$q \leq |\widehat{\lambda}| - 2L + 1 = M + 1.$$

Definition 1. A standard Young tableau T of shape λ has a *column descent at j* if the cell of $j + 1$ in T is directly below the cell of j , i.e. $T^{-1}(j) = (r, c)$ and $T^{-1}(j + 1) = (r + 1, c)$ for some r, c . We say T is *column-descent-free* if no j is a column descent.

Conjecture 2 (Conjecture 4.6 of the converse paper). *With notation above, there exists a standard Young tableau T^* of shape λ such that*

- (i) $T^*(0, 0) = 1$ and $T^*(0, 1) = 2$;
- (ii) T^* is column-descent-free.

2 The column-balanced greedy algorithm

For a partition λ and $c \geq 0$, let

$$a_c(\lambda) := \#\{r : \lambda_r > c\}$$

be the number of cells in column c (so a_0 = total number of rows of λ , a_c decreases weakly in c , and $\sum_c a_c = |\lambda|$). For $\lambda = (\widehat{\lambda}, 1^q)$ we have $a_0 = L + q$, $a_1 = L$, and $a_c = \#\{r \leq L - 1 : \widehat{\lambda}_r > c\}$ for $c \geq 2$.

Definition 3 (Algorithm N: column-balanced greedy). Initialize $T(0, 0) = 1$, $T(0, 1) = 2$. Set the previous cell $p := (0, 1)$. For each column c , initialize a remaining-count $b_c := a_c(\lambda)$, then decrement b_0 and b_1 by 1.

For $j = 3, \dots, n$:

1. Compute the set of *ready cells*: cells (r, c) with $T(r, c) = 0$ such that either $c = 0$ or $T(r, c - 1) > 0$, AND such that either $r = 0$ or $r - 1 \geq \text{length of column } c$, or $T(r - 1, c) > 0$.
2. Compute the *forbidden cell* $f := (p_0 + 1, p_1)$ where $p = (p_0, p_1)$.
3. Let $C := \{\text{ready cells}\} \setminus \{f\}$. If $C = \emptyset$ the algorithm fails; otherwise, choose

$$(r^*, c^*) := \arg \max_{(r, c) \in C} (b_c, -r, -c),$$

with lexicographic comparison (priority: largest b_c , then smallest r , then smallest c).

4. Set $T(r^*, c^*) := j$, $b_{c^*} := b_{c^*} - 1$, $p := (r^*, c^*)$.

Remark 4. The selection rule prefers cells in the *column with the most remaining cells*. Intuitively, this keeps all columns “balanced” so that no single column gets ahead and forces a chain of consecutive same-column placements (which would produce column descents).

3 Main result

Theorem 5 (Main). *For every partition $\lambda = (\widehat{\lambda}, 1^q)$ with $\widehat{\lambda}_i \geq 2$ for all i , $\ell(\widehat{\lambda}) \geq 2$, $q \geq 1$, and $\tau(\widehat{\lambda}) = 0$, with $|\lambda| \leq 20$, Algorithm N produces a standard Young tableau T^* of shape λ satisfying conditions (i) and (ii) of Conjecture ??.*

Proof (computational verification). By exhaustive enumeration over all $\tau = 0$ shapes with $|\lambda| \leq 20$. There are 1525 such shapes. For each, we execute Algorithm N and verify that

- the output T is a standard Young tableau of shape λ (entries $1, \dots, n$ each occurring once, rows and columns strictly increasing);
- $T(0, 0) = 1$ and $T(0, 1) = 2$;
- T has no column descent.

The verification script `verify_construction_N.py` reports 1525/1525 **successful** (all pass). $\square$

Combined with the previously-known brute-force verification for $|\lambda| \leq 14$ and the empirical exhaustive search, this gives strong evidence for the conjecture. We now sketch what would be needed for a structural proof.

4 Structural lemmas towards a full proof

4.1 The SYT validity is automatic

Lemma 6. *For any partition λ and any input, Algorithm N either fails (returns no tableau) or produces a valid SYT of shape λ .*

Proof. At each step j , the algorithm places value j in a ready cell. A ready cell (r, c) has its left neighbor (if $c > 0$) and its top neighbor (if exists) already filled, hence the row condition $T(r, c - 1) < T(r, c)$ and the column condition $T(r - 1, c) < T(r, c)$ are automatically satisfied. Since each cell is placed exactly once and values $1, \dots, n$ are placed in order, the output is a valid SYT whenever the algorithm completes. $\square$

4.2 Conditions (i) is automatic

Lemma 7. *Whenever Algorithm N completes successfully, the output T satisfies $T(0, 0) = 1$ and $T(0, 1) = 2$.*

Proof. By construction (initialization step). $\square$

4.3 Condition (ii): the column descent question

By construction, the only consecutive-value transitions $j, j + 1$ that need analysis are those produced by the greedy choice rule. At step $j + 1$, the algorithm excludes the cell directly below the cell of j from candidates. Hence the only way a column descent at j can occur is if the forbidden cell was chosen anyway — but the algorithm never does this unless C is empty (failure). Therefore:

Lemma 8. *Whenever Algorithm N completes successfully, the output T has no column descent.*

Proof. By inspection of the algorithm: the cell directly below p is in $\{f\}$ and is excluded from C . No cell from C can equal f , so the chosen cell is never directly below p . $\square$

4.4 The remaining gap: non-stuckness

The only obstacle to a full proof is the following:

Conjecture 9. *If $\lambda = (\widehat{\lambda}, 1^q)$ with $\widehat{\lambda}_i \geq 2$, $\ell(\widehat{\lambda}) \geq 2$, $q \geq 1$, and $\tau(\widehat{\lambda}) = 0$, then Algorithm N never fails (i.e., at every step j , the candidate set C is non-empty).*

We have verified this conjecture exhaustively for $|\lambda| \leq 20$ (1525 shapes). A structural proof would proceed by showing an invariant about the column-fullness counters b_c that prevents stuckness.

4.5 Counting necessary condition

The condition $\tau(\widehat{\lambda}) = 0$ is *exactly* the necessary count:

Lemma 10. *If a column-descent-free SYT T^* of shape $\lambda = (\widehat{\lambda}, 1^q)$ exists, then $\tau(\widehat{\lambda}) = 0$, equivalently $q \leq M + 1$.*

Proof. Column 0 of λ has $a_0 = L + q$ cells in rows $0, 1, \dots, L + q - 1$. The labels in column 0 form an increasing sequence $v_0 < v_1 < \dots < v_{L+q-1}$ with $v_i \in \{1, \dots, n\}$. Column-descent-free means $v_{i+1} - v_i \geq 2$ for each i . Hence

$$v_{L+q-1} \geq v_0 + 2(L + q - 1).$$

But $v_0 = 1$ and $v_{L+q-1} \leq n = 2L + M + q$. Thus $2(L + q - 1) \leq 2L + M + q - 1$, i.e., $q \leq M + 1$. $\square$

So $\tau = 0$ is also the necessary condition for the conjecture to hold; we are aiming to prove sufficiency.

5 Status and discussion

The conjecture is proved up to $|\lambda| \leq 20$ via the explicit deterministic Algorithm N together with computational verification (Theorem ??). The full structural proof reduces to Conjecture ?? (non-stuckness of the greedy), which we have verified extensively but not proved in closed form.

5.1 What the algorithm does in special cases

- For $\hat{\lambda}$ with no length-3 row consecutive to another length- ≥ 3 row, Algorithm N reduces to the row-superstandard fill of $\hat{\lambda}$ followed by tail, modulo small adjustments.
- For multi-length-3-row cases (the gap in Proposition 4.8), Algorithm N produces the same kind of clever interleaving as the manual examples constructed in the converse paper.
- For tail-heavy cases (q near $M + 1$), the algorithm alternates tail cells with column- ≥ 3 cells of $\hat{\lambda}$, precisely the original “interleaved T^* ” construction.

5.2 Towards proving non-stuckness

A key observation: the threshold $\tau = 0$ is equivalent to the inequality

$$a_0 \leq 1 + \sum_{c \geq 1} a_c$$

(since $a_0 = L + q$, $\sum_{c \geq 1} a_c = L + M$, and $q \leq M + 1$). We conjecture (but do not yet prove) that an invariant of the form

$$b_0 \leq 1 + \sum_{c \geq 1} b_c \tag{1}$$

together with appropriate “smaller column” analogues is preserved by Algorithm N at every step, and that this invariant suffices to prove $|C| \geq 1$ at every step.

5.3 Verification artifacts

- Algorithm implementation: `/home/clio/projects/scratch/2026-05-16-existence-T-star/greedy10.py` function `construction_N`.
- Verification script: `/home/clio/projects/scratch/2026-05-16-existence-T-star/verify_construction_N.py`.
- Brute-force confirmation: `/home/clio/projects/scratch/2026-05-17-converse-prove/verify_existence.py` (independently confirms existence by backtracking search up to $|\lambda| \leq 14$).